

Week 2 Task 8 — Cyber Kill Chain + MITRE ATT&CK; Intro

Cybersecurity Internship Project Report

This project introduces the Lockheed Martin Cyber Kill Chain and MITRE ATT&CK;, focusing on T1110 — Brute Force.

1. Cyber Kill Chain — Seven Stages

Stage	Description
1. Reconnaissance	Research, identify, and select targets.
2. Weaponization	Prepare a malicious payload or deliverable.
3. Delivery	Transmit the payload to the target.
4. Exploitation	Trigger malicious code through a vulnerability or user action.
5. Installation	Establish a malicious presence on the target system.
6. Command & Control	Establish communication for adversary control.
7. Actions on Objectives	Perform the intended mission, such as theft or disruption.

2. MITRE ATT&CK; — T1110 Brute Force

T1110 is a Credential Access technique involving systematic credential attempts. MITRE lists Password Guessing, Password Cracking, Password Spraying, and Credential Stuffing as sub-techniques.

3. Practical Evidence

Original MITRE ATT&CK; screenshot supplied for this project:

The screenshot displays the MITRE ATT&CK framework website. The top navigation bar includes links for Matrices, Tactics, Techniques, Defenses, CTI, Resources, Benefactors, Contribute, and Blog. A search bar is located below the navigation bar. On the left side, a sidebar lists various technique categories: Enterprise, Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense, Impairment, Credential Access, Discovery, and Lateral Movement. The main content area shows a table of techniques. The table has columns for ID, Name, and Description. The techniques listed are:

ID	Name	Description
T1185	Browser Session Hijacking	Adversaries may take advantage of security vulnerabilities and inherent functionality in browser software to change content, modify user-behaviors, and intercept information as part of various browser session hijacking techniques.
T1110	Brute Force	Adversaries may use brute force techniques to gain access to accounts when passwords are unknown or when password hashes are obtained. Without knowledge of the password for an account or set of accounts, an adversary may systematically guess the password using a repetitive or iterative mechanism. Brute forcing passwords can take place via interaction with a service that will check the validity of those credentials or offline against previously acquired credential data, such as password hashes.
.001	Password Guessing	Adversaries with no prior knowledge of legitimate credentials within the system or environment may guess passwords to attempt access to accounts. Without knowledge of the password for an account, an adversary may opt to systematically guess the password using a repetitive or iterative mechanism. An adversary may guess login credentials without prior knowledge of system or environment passwords during an operation by using a list of common passwords. Password guessing may or

4. Attack Mapping — Phishing-Based Malware Scenario

Stage	Example Activity
Reconnaissance	Identify target and gather information.
Weaponization	Prepare malicious document or payload.
Delivery	Send phishing message.
Exploitation	Victim opens malicious content and attack triggers.
Installation	Malware establishes a presence.
Command & Control	Compromised host communicates with attacker infrastructure.
Actions on Objectives	Attempt theft, disruption, or another intended objective.

5. Defensive Detection Opportunities

Email security, URL filtering, endpoint monitoring, patch management, network monitoring, audit logging, access controls, and behavioral analytics can help detect or disrupt activity across the attack lifecycle.

6. SOC Analyst Relevance

The Cyber Kill Chain provides a high-level sequence of an intrusion, while MITRE ATT&CK; provides detailed adversary behaviors. Repeated authentication failures can help analysts investigate possible T1110 activity.

7. Conclusion

This project establishes a practical foundation for understanding attack lifecycles and mapping adversary behavior to MITRE ATT&CK; techniques.

Evidence Integrity: the included screenshot is the original image supplied for this project. No fabricated screenshot evidence has been added.